UNIVERSITY OF CINCINNATI · INNOVATION CHALLENGE

Reimagining Concussion Management for Young Athletes



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THE PROBLEM

Current Concussion Protocols Fall Short

150K+

annual HS football concussions

Eyewitness Only

Diagnosis relies on human observation — invisible injuries are routinely missed

No Quantifiable Data

No objective measurements to support return-to-play decisions

CTE Risk

Repetitive mild TBI can lead to chronic traumatic encephalopathy (CTE)

Secondary Injury

Non-immediate effects of injury are consistently overlooked on the field

THE SOLUTION

Designed to Protect, Not Interrupt

01

Increased Visibility

Makes potential concussions immediately visible on the field without relying solely on eyewitness accounts.

02

Near-Immediate Alerts

Real-time notification system alerts coaches and medical staff within seconds of a high-impact event.

03

Zero Game Disruption

Integrated seamlessly into existing equipment, athletes play without interference, and coaches get the information they need.

H.I.T.T.

Helmet · Integrated · TBI · Tracker



Microcontrollers

- Arduino-based system
- Outputs real-time data and alerts
- Force sensors for impact
- Miniaturized for helmets



Helmet Integration

- Fits existing equipment
- Position-specific design
- Easy setup & removal
- No performance impact



Live Data Interface

- Real-time readouts
- Statistical analysis
- Sideline dashboard
- Historical tracking

THE DEMO

H.I.T.T. in Action



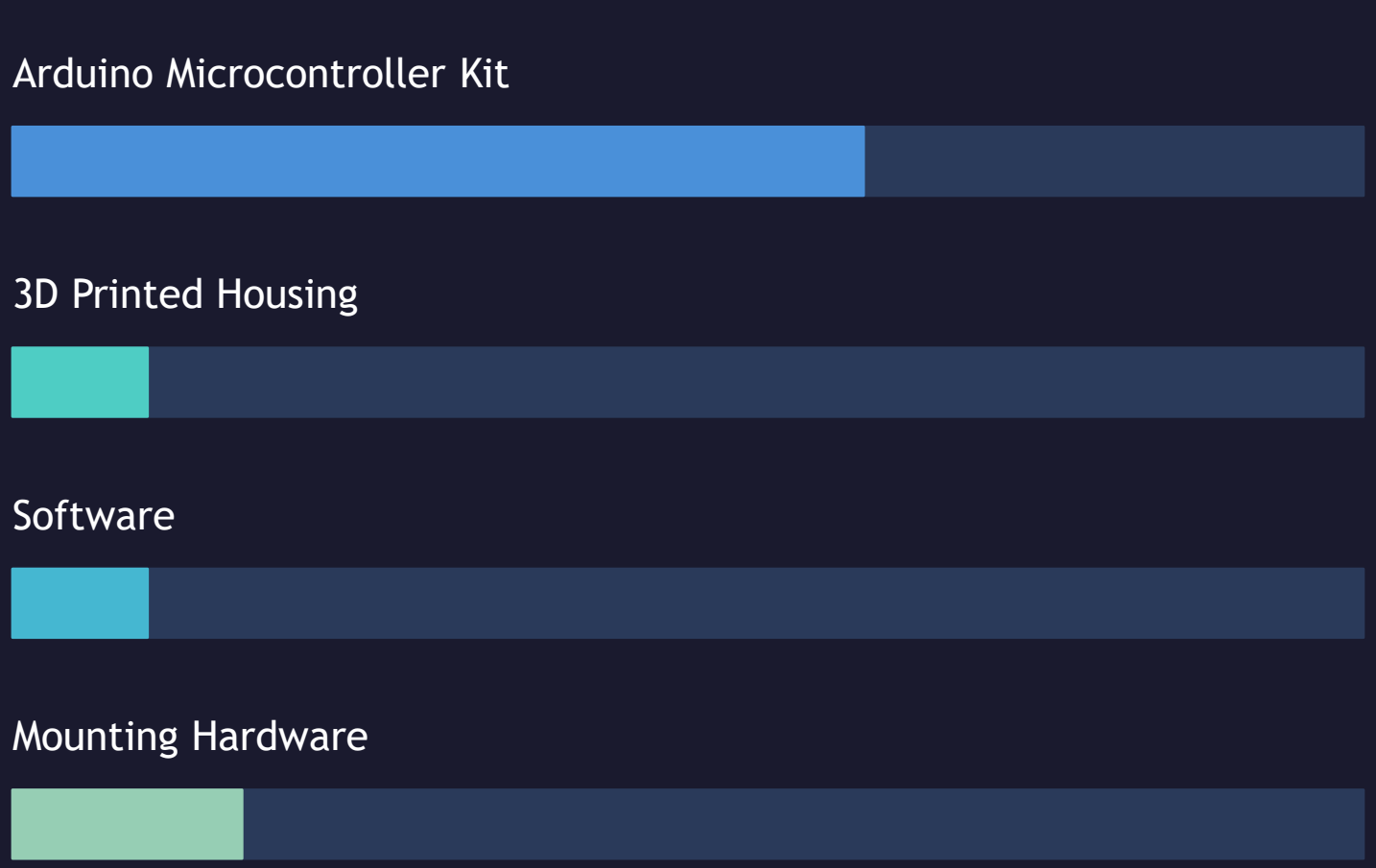
COMPETITIVE LANDSCAPE

Why H.I.T.T. Wins

Feature	ACT Band	Riddell InSite	DASH System	H.I.T.T.
Football-Specific	✗	✓	✓	✓
Removable	✓	✗	✗	✓
Affordable (<$100)	✓	✗	✗	✓
No Subscription	✓	✗	?	✓
Live Sideline Data	✗	✓	✓	✓

MANUFACTURING COST

Built to Be Affordable



TOTAL COST

$48

per unit

vs. $999 competitor

95% less expensive

BUSINESS MODEL

How We Make Money

PRIMARY · HARDWARE SALE

$99 / unit

Sell directly to high school programs. A 50-player roster = $4,950

UNIT ECONOMICS

Sell price	$99
Cost to make	$48
Gross margin	$51
Margin %	51.5%

REVENUE PROJECTION

	Programs	Units Sold	Revenue
Year 1	20 schools	1,000 units	$51,000
Year 2	100 schools	5,000 units	$255,000
Year 3	400 schools	20,000 units	$1.02M

MARKET OPPORTUNITY

A Large, Underserved Market

Total Concussion & Smart Helmet Market	Football Helmet Sensor Market	H.I.T.T. Serviceable Target (HS Football)
$7.5B	$892M	$121M

14,000+	1.1M+	~$4,950	51.5%
High School Football Programs	HS Football Players Annually	Cost per Team at $149/unit	Gross Margin per Unit

THE ASK

Here's What We Need to Win

40%

Product Development

Miniaturize from Arduino prototype to custom PCB. Reduce per-unit cost from $48 to ~$28 at scale.

35%

Pilot Programs

Partner with 5–10 local high school programs for real-world testing, data collection, and case studies.

25%

Regulatory & IP

File provisional patent. Begin FDA Class II medical device pathway for longer-term credibility.

We have a working prototype, real coach validation, a $121M market, and a cost structure that no competitor can match.

THE TEAM

Thank You - Any Questions?

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